

Automated Acronym Understanding and Hierarchical Concept Mapping for Scientific Text: From Research Design to Prototype Implementation (HMRP)

Research Project Draft — Compiled from Project Documentation

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1 Abstract

Scientific and technical documents rely heavily on acronyms and abbreviations, and automatically identifying and correctly interpreting these short forms remains an open problem in Natural Language Processing (NLP). This paper synthesizes two linked bodies of work produced within the same project: (1) a research design phase (Phases 1–6 of the project, extended by a 22-paper consolidated literature comparison) that reviews the state of the art in **Acronym Identification (AI)** and **Acronym Disambiguation (AD)** and proposes a ten-module pipeline that extends flat acronym resolution into a **Concept-Category-Context (CCC)** hierarchical knowledge representation with an interactive visualization layer; and (2) a working prototype, referred to in the project artifacts as **HMRP (Hierarchical Concept Mapping for Research Paper Similarity Analysis)**, which implements PDF ingestion, multi-engine keyword/concept extraction (TF-IDF, KeyBERT, YAKE, spaCy NER), a fine-tuned SciBERT-based acronym identification and sequence-classification model, domain-taxonomy-driven hierarchy construction, an interactive knowledge-graph/visualization layer, and dual-engine (lexical + dense-embedding) paper similarity analysis. The prototype was evaluated on a curated, leakage-free 50-paper arXiv corpus (1,123 pages, document-level 70/14/16 train/validation/test split). Reported results show a measured increase in extracted concepts per paper from a 15-concept unigram baseline to 60 concepts under the multi-engine pipeline, 100% page/sentence evidence traceability (versus 0% for the baseline), and a 0% hallucination rate as enforced by an 8-rule paper validator, alongside pairwise semantic similarity statistics on the test partition (mean cosine similarity 0.2424, standard deviation 0.0667). We map the implemented modules explicitly against the originally proposed ten-module CCC architecture, showing that Modules 1–6 (PDF processing through disambiguation-oriented sequence classification) and a simplified version of Modules 9–10 (domain-taxonomy hierarchy and knowledge-graph visualization) are realized in the prototype, while full CCC chain construction, typed semantic-relationship extraction, and formal benchmarking against the SciAI/SciAD gold-standard datasets remain unimplemented and are identified as concrete future work. This paper reports only what is evidenced in the project’s own documentation, code structure, dataset reports, and accuracy reports, and explicitly marks unverified or unmeasured claims as *Not Available*.

2 Keywords

Acronym Identification, Acronym Disambiguation, Scientific Document Understanding, SciBERT, Concept-Category-Context (CCC) Mapping, Hierarchical Knowledge Representation, Knowledge Graph, Keyword Extraction, KeyBERT, TF-IDF, Semantic Similarity, Scientific NLP, Literature Review Automation

3 Introduction

Scientific writing makes pervasive use of acronyms and technical abbreviations. Correctly detecting a short form (e.g., “CNN”), correctly resolving its full form when several candidate meanings exist, and — beyond that — organizing the resolved terms into a structure a reader can navigate, are three progressively harder problems. A large body of NLP research (surveyed in Section 9) has addressed the first two problems — Acronym Identification (AI) and Acronym Disambiguation (AD) — largely as flat, sentence-level classification tasks, pushing benchmark accuracy from the mid-80s to the low-to-mid-90s (Macro-F1) against a human-performance ceiling of roughly 96%. However, as documented in this project’s own literature review (Phases 1-3) and its extension to 22 papers (the *Research Blueprint*), no reviewed system organizes resolved acronyms into a higher-level, explorable knowledge structure.

This project proceeded in two stages, both of which are documented in the uploaded materials and are reported here without alteration of their substance:

1. **A research design stage** (Phase1_Problem_Understanding.md through Phase6_Algorithm_Model_Design.md, consolidated in Project_Report_Phases_1-6.pdf, and extended in Research_Blueprint_Report.pdf/Consolidated_Paper_Comparison.xl that reviews six core AI/AD papers (later extended to 22 papers across four themes), identifies a research gap, and proposes a ten-module pipeline culminating in Concept-Category-Context (CCC) mapping, hierarchy generation, knowledge-graph construction, and interactive visualization.
2. **An implementation stage**, documented in Finalreport.md, DATASET_REPORT.md, and MODEL_ACCURACY_REPORT.md, describing a working system referred to as **HMRP** — a Streamlit-based research-paper analysis platform that ingests PDFs, extracts keywords/concepts through a multi-engine pipeline, applies a fine-tuned SciBERT model for acronym-related token and sequence classification, builds a domain-taxonomy hierarchy and knowledge graph, computes pairwise paper similarity, and enforces an 8-rule validation protocol against data leakage and hallucination.

This paper’s purpose is to present these two stages as a single, honest research narrative: what was proposed, what was actually built and measured, and where the two diverge. Wherever the implementation stage does not provide direct evidence for a claim made in the design stage, that gap is stated explicitly rather than assumed to be closed.

4 Problem Statement

Scientific and technical documents make heavy use of acronyms and abbreviations, but automatically recognizing these short forms and correctly resolving what they mean is still an unsolved problem for NLP systems. Two coupled sub-problems exist:

- **Acronym Identification (AI):** given a sentence, detect which spans are acronyms (short forms) and which spans are their corresponding long forms (definitions).

- **Acronym Disambiguation (AD):** given an acronym that has multiple possible expansions (e.g., “CNN” → *Convolutional Neural Network* or *Cable News Network*), determine the correct long form based on surrounding context.

Historically, research on these tasks was limited by datasets that were too small, generated through noisy rule-based heuristics, or narrowly confined to the medical/biomedical domain. Even after large benchmark datasets (SciAI, SciAD) were introduced, subsequent work revealed additional problems — annotation noise, duplicate examples with conflicting labels, and a persistent performance gap between the best models (~81-94% Macro-F1) and human-level accuracy (~96% F1).

Beyond identification and disambiguation, a second, largely unaddressed problem motivates this project: once an acronym is correctly resolved, no reviewed system organizes the result into a structured, explorable representation of how the underlying concept relates to other concepts in the same document or corpus. Separately, at the systems level, researchers surveying large PDF corpora face information overload, reliance on surface keyword search, and risk of hallucinated or cross-document-contaminated output from automated summarization tools — a practical problem that the implemented HMRP prototype targets directly.

5 Motivation

- Acronyms are pervasive in technical writing: roughly 15% of PubMed queries and a large share of scientific/clinical text contain abbreviations.
- Misinterpreting an acronym can propagate errors into downstream NLP applications such as question answering, information retrieval, and definition extraction.
- Acronyms are frequently ambiguous, and their correct meaning often depends on distant contextual cues rather than immediately adjacent words.
- Prior benchmark datasets (SciAI, SciAD) were shown to contain noise, duplication, and conflicting labels, which both limits model training and biases reported evaluation results.
- No unified, publicly available, multi-domain tool existed (as of the reviewed literature) that both resolves acronyms and organizes the result into structured, explorable knowledge.
- Independently, researchers reviewing dozens of papers at once need automated tools that avoid keyword-only search, provide verifiable (non-hallucinated) evidence for every extracted claim, and prevent information from one paper from leaking into the analysis of another — the practical motivation behind the HMRP prototype’s evidence-grounding and 8-rule validation design.

6 Research Objectives

6.1 Objectives from the Research Design Phase (Phases 1-4)

1. Build/use large-scale, high-quality, human-annotated datasets for acronym identification and disambiguation in the scientific domain.

2. Design or apply deep learning models that identify acronym-long form pairs in text.
3. Design or apply models that disambiguate acronyms with multiple candidate meanings using sentence-level context.
4. Evaluate and benchmark these models using standard metrics (Precision, Recall, Macro-F1) against existing baselines and human performance.
5. Identify and, where possible, correct data-quality issues that bias evaluation.
6. Explore CCC (Concept-Category-Context) mapping, hierarchy generation, knowledge-graph construction, and interactive visualization as a novel extension beyond flat acronym resolution.
7. (Optional) Package the approach into a usable, deployable system.

6.2 Objectives Realized in the Implementation Stage (HMRP)

1. Automated PDF parsing: recursively scan, ingest, and parse academic PDFs, extracting text, sections, metadata.
2. Multi-engine concept/keyword extraction merging statistical, contextual-embedding, n-gram, and linguistic-entity extractors.
3. Domain-specific scientific term identification via a fine-tuned SciBERT model.
4. Hierarchical taxonomy generation mapping extracted concepts to broad research domains.
5. Multi-modal pairwise similarity computation to cluster related papers.
6. Strict paper isolation and leakage-free validation (8 mandatory rules).
7. Full source traceability linking every extracted keyword/concept to a verbatim sentence and page number.
8. Interactive web dashboard and multi-format export.

Note on alignment: Objectives 4.2(1)–4.2(3) and 4.2(6)–4.2(7) are directly evidenced in the implementation documentation. Objective 4.1(6) — CCC mapping, hierarchy, knowledge graph, and visualization as a *dedicated, per-acronym recursive structure with typed semantic edges* — is only **partially** realized: the implemented system provides a domain-taxonomy hierarchy and a knowledge graph, but not the specific CCC chain-construction algorithm or the typed relationship-extraction step (used_for, is_a, part_of, etc.) described in the design phase. This distinction is maintained throughout the paper and revisited in Sections 15, 17, 18, and 20.

7 Research Gap

Two literature reviews were conducted within this project: an initial six-paper review (Phase 2) and an extended, 22-paper consolidated comparison spanning four themes (the *Research Blueprint*). Both converge on the same gap statement.

7.1 Six-Paper Gap Statement (Phase 2 / Phase 3)

Existing studies on scientific acronym understanding focus almost entirely on two flat, sentence-level sub-tasks: identifying acronym/long-form spans, and disambiguating an acronym’s correct meaning from a set of candidates. State-of-the-art methods

— spanning syntactic graph models (GAD), adversarially-trained transformer ensembles (AT-BERT), rule-based multi-domain systems (MadDog), multi-strategy binary classifiers (Pan et al.), dual-path domain-fused transformers (hdBERT), and retrieval-based disambiguation (Primer AI) — push accuracy close to human level (~91-94% F1 vs. ~96% human). None of these systems organize resolved acronyms into higher-level concepts, establish semantic relationships between them, build a hierarchical or graph-based knowledge structure, or provide an interactive way for a reader to explore how the concepts in a paper relate to one another.

7.2 22-Paper Extended Gap Statement (Research Blueprint)

Extending the review to 22 papers across four themes — (A) Scientific Acronym ID/AD (6 papers), (B) Clinical/Biomedical Abbreviation Detection/Disambiguation (8 papers), (C) Knowledge Graph / Concept Map Construction (6 papers), and (D) Taxonomy / Hierarchy Construction (3 papers) — confirms and sharpens the gap:

- **Themes A and B (14 papers)** solve identification/disambiguation but stop at a flat resolved label; none link the output onward into a concept, category, or hierarchy.
- **Themes C and D (9 papers, with Graphusion bridging A/C)** solve concept/knowledge organization but never start from a resolved abbreviation — they extract concepts from raw text or a pre-given term list.
- **No paper in either group performs CCC-style (Concept-Category-Context) mapping**, i.e., an explicit category assignment plus per-concept sentence/document provenance chained upward from a resolved acronym.
- **No paper combines interactive visualization with abbreviation resolution** end-to-end; MadDog offers direct lookup only, and the concept-map papers (GT-D2G, Automated Concept Map Extraction) are disconnected from any abbreviation pipeline.

Research Gap Statement: *No single reviewed system takes an abbreviation from raw text all the way through disambiguation into a structured, hierarchical, and visually explorable concept representation, with each layer’s evidence traceable to source text.*

This gap statement is the basis for the proposed ten-module methodology (Section 10) and is the framing against which the implemented prototype (Sections 11-17) is evaluated.

8 Related Work / Literature Review

8.1 Core Review: Acronym Identification and Disambiguation (6 Papers)

#	Paper (Authors, Year, Venue)	Method	Dataset	Reported F1	Key Limitation
1	Veyseh et al., 2020, COLING (arXiv:2010.14578)	LSTM-CRF (AI); GAD = BiLSTM+GCN over dependency tree (AD)	SciAI (17,506 sent.), SciAD (62,441 samples)	LSTM-CRF 86.55 (AI); GAD 81.90 (AD); human $\approx$ 96.1	Well below human performance; annotation noise
2	Zhu et al., 2021, AACL 2021, SDU (arXiv:2101.03700)	Fine-tuned BERT/SciBERT + FGM adversarial training + ensembling	SciAI (17,506 sent.), SciAD (62,441 samples)	AT-BERT-E 91.52 (AI); BERT-E 91.52 (AI); BERT-E 91.52 (AI)	$\sim$ 2x training cost; still below human
3	Veyseh et al., 2021, EACL 2021 Demos (arXiv:2101.09881)	Rule-based AI (extended Schwartz-Berst) + BiLSTM AD (DOG glossary + MAD dataset)	DOG (426,389 acronyms), MAD (46M records)	88.12 (AI); 88.49 (AD)	64 GB storage; no visualization beyond lookup
4	Pan et al., 2021, AACL 2021, SDU (arXiv:2103.00488)	SciBERT binary classifier + dynamic negative sampling + TAPT + adversarial training + pseudo-labeling	SciAD (62,441 samples)	P 0.9695 / R 0.9132 / F1 0.9405	Struggles with near-synonymous candidates
5	Zhong et al., 2021, AACL 2021, SDU (arXiv:2107.00816)	hdBERT: dual-path RoBERTa + SciBERT $\rightarrow$ SILE	SciAD_BI (352K/28K/28K)	93.73	Below human recall; sensitive to noisy labels

#	Paper (Authors, Year, Venue)	Method	Dataset	Reported F1	Key Limitation
6	Egan & Bohannon, 2021, AAAI-21 SDU (arXiv:2012.08043)	XLNet ensemble (AI); Siamese/Twin retrieval on a 13 (AD)	SciAI, SciAD, AuxAI, AuxAD, SciAD- dedupe	92.60 (AI); 91.58 (AD)	Retrieval fails without similar examples; found 12,672 dupli- cate/conflicting SciAD labels

Common thread: every paper improves accuracy on a flat, sentence-level task. None asks what happens *after* an acronym is correctly resolved.

8.2 Extended Review: 22 Papers Across Four Themes (Research Blueprint)

The consolidated comparison spreadsheet (Consolidated_Paper_Comparison.xlsx) extends the review with 16 additional papers, grouped into three further themes:

Theme B — Clinical/Biomedical Abbreviation Detection & Disambiguation (8 papers): Zilio et al. 2022 (PLOD, 160K+ segments, LREC); Wagh & Khanna 2023 (ClinicalBERT vs. SciBERT vs. BioBERT on CASI, best F 91.49%); Wen, Lu & Reddy 2020 (MeDAL, 14.3M PubMed articles, a pretraining resource); Skreta et al. 2021 (ontology-aware CNN, +17% via biomedical-ontology augmentation, Nature Communications); JMIR 2024 (One-to-All BlueBERT, 95–98% accuracy on MSH WSD/UMN, but truncated by BERT’s 512-token limit); Adams et al. 2020 (Latent Meaning Cells, zero-shot clinical acronym expansion); a 2024 PubMed/JAMIA-adjacent study (zero-shot LLM clinical acronym disambiguation).

Theme C — Knowledge Graph / Concept Map Construction (6 papers): Zhong et al. 2023 (ACM Computing Surveys — a 300+ method KG-construction survey with a 3-step acquisition/refinement/evolution pipeline); Yang et al. 2024 (*Graphusion*, arXiv:2407.10794 — zero-shot LLM-based KG construction and fusion for NLP-education content, with a TutorQA benchmark); the 2025 *NLP-AKG* paper (arXiv:2502.14192 — a large-scale academic KG over ACL Anthology, 620,353 entities, 2,271,584 relations); Sahrawat et al. 2019/2020 (SciBERT-based scientific keyphrase extraction on ScienceIE); GT-D2G (2022, NAACL Findings — task-guided neural concept-map generation); *Automated Concept Map Extraction from Text* (2024, OpenReview — an open-source modular concept-map system, METEOR 25.69).

Theme D — Taxonomy / Hierarchy Construction (3 papers): Mao et al. 2018 (*TaxoRL*, ACL — reinforcement-learning taxonomy construction); Zhang et al. 2018

(*TaxoGen*, KDD — unsupervised clustering-based topic taxonomy); Shang et al. 2018 (*HiExpan*, KDD — seed-taxonomy expansion).

Cross-theme observation: Themes C and D provide general-purpose tools for turning text or a term list into a structured concept hierarchy or graph, but none of the 9 papers in these themes take a *resolved abbreviation* as their starting node, and none combine this with abbreviation resolution end-to-end. Graphusion is the closest analog (concept/relation extraction for NLP-domain text) but is not abbreviation-aware and carries LLM-hallucination risk in its triplet-extraction step.

8.3 Chronological Evolution (Six Core Papers)

The field moved from (a) dataset construction and a first syntactic baseline (Veyseh et al., 2020) → (b) transformer fine-tuning with robustness tricks (AT-BERT, 2021) → (c) task reframing as binary classification / IR-based disambiguation (Pan et al.; Egan & Bohannon, 2021) → (d) domain-knowledge fusion (hdBERT, 2021) → (e) auditing and correcting benchmark data quality itself (Egan & Bohannon’s SciAD-dedupe). By late 2021, both AI (~92-94% F1) and AD (~91-94% F1) had closed most, but not all, of the gap to human performance (~96% F1); remaining bottlenecks shifted from raw model capacity toward data quality and hard residual cases (long-distance dependency, near-synonymous candidates).

9 Proposed Methodology

The methodology below is the design produced in Phase 4 (“Proposed Methodology”) and detailed algorithmically in Phase 6 (“Algorithm & Model Design”). It is presented here as the **proposed** target architecture; Section 11 onward reports which parts were actually implemented.

9.1 Input and Output Specification

- **Input:** a research paper as `paper.pdf`, or raw extracted text.
- **Output (proposed, full chain):** abbreviation list → disambiguated full forms → concept categories → CCC mapping (Concept-Category-Context) → hierarchical tree → knowledge graph → interactive visualization.

Example output chain (illustrative, from the design documents):

CNN → Convolutional Neural Network → Deep Learning → Machine Learning → Artificial Intelligence

9.2 Ten-Module Design

#	Module	Grounded in prior work	Status
1	PDF Processing	Common preprocessing across Papers 1–6	Implemented
2	Text Extraction	Common preprocessing across Papers 1–6	Implemented
3	Preprocessing (cleaning, segmentation)	Common preprocessing across Papers 1–6	Implemented
4	Abbreviation Detection (BIO tagging)	Papers 1 (LSTM-CRF), 2 (AT-BERT), 6 (XLNet ensemble)	Implemented (SciBERT token classifier)
5	Long Form Extraction	Papers 1, 3 (MadDog rule-based + DOG glossary)	Implemented (rule-based/dictionary, via concept extractor)
6	Disambiguation	Papers 1, 3, 4, 5, 6	Partially implemented (SciBERT sequence classifier; not a full multi-candidate argmax disambiguator)

#	Module	Grounded in prior work	Status
7	Concept Extraction	<i>New — not covered in reviewed literature</i>	Partially implemented (domain-taxonomy keyword matching, not embedding-anchor concept mapping as specified)
8	CCC Mapping	<i>New — core proposed contribution</i>	Not implemented as a discrete recursive chain-construction algorithm
9	Hierarchy Generation	<i>New</i>	Implemented (domain-taxonomy parent-child tree)
10	Knowledge Graph + Visualization	<i>New</i>	Implemented (NetworkX + PyVis + Plotly), without typed semantic-relation edges from Module 8b

9.3 Module-Level Algorithm Summaries (as designed)

- **Module 4 (Abbreviation Detection):** SciBERT encoder + linear BIO classification head; optional FGM adversarial perturbation during fine-tuning (following AT-BERT). Fallback: rule-based candidate detection (all-uppercase tokens, length 2-6, in parentheses).
- **Module 5 (Long Form Extraction):** Parenthesis rule and Schwartz & Hearst initialism alignment, with dictionary lookup fallback and Levenshtein-distance normalization.
- **Module 6 (Disambiguation):** Proposed as an hdBERT-style dual-path fusion (RoBERTa + SciBERT $\rightarrow$ MLP sigmoid), layered with Pan et al.’s training strategies (dynamic negative sampling, TAPT, adversarial training, pseudo-labeling); retrieval-based fallback (Siamese network, nearest-neighbor label copying) when no fixed candidate set exists.
- **Module 7 (Concept Extraction, novel):** Dictionary lookup against a curated AI/CS taxonomy, with an embedding-similarity fallback (cosine similarity against concept-anchor terms) below a confidence threshold, else flagged for manual labeling.
- **Module 8 (CCC Mapping, novel/core):** `ccc_map(abbreviation, full_form)` recursively walks a category taxonomy from the resolved concept upward to a root concept (e.g., Artificial Intelligence), retaining the originating sen-

tence/section as “Context” provenance.

- **Module 9 (Hierarchy Generation):** Parent-child adjacency construction from CCC chains; $O(n \cdot d)$ construction, $O(V)$ traversal.
- **Module 8b/10 (Relationship Extraction + Knowledge Graph):** Pattern-matching over a fixed relation vocabulary (used_for, belongs_to, part_of, extends, improves, depends_on, similar_to), with a transformer-based fallback classifier; graph assembled with NetworkX.
- **Module 10b (Visualization):** Tree, force-directed graph, or radial mind-map rendering (PyVis / Cytoscape.js / D3.js), with click-to-expand and hover-for-context interactivity.

9.4 Technology Stack (as proposed)

Module	Suggested Technology
PDF Processing	PyMuPDF, pdfplumber
NLP (cleaning, tokenization)	spaCy, SciSpaCy
Transformer backbone	SciBERT, BERT
Abbreviation detection	SciAI-based sequence labeling or rule-based
Disambiguation	SciBERT / hdBERT-style dual-path model
Concept/relationship & graph processing	NetworkX
KG storage (optional, at scale)	Neo4j
Visualization	PyVis, D3.js, Cytoscape

9.5 Proposed Evaluation Plan (Two Tiers)

- **Tier 1 (Modules 1-6):** benchmarked against Papers 1-6 using Precision/Recall/Macro-F1 on SciAI/SciAD/SciAD-dedupe, with ~96% human performance as the ceiling reference.
- **Tier 2 (Modules 7-10):** evaluated via expert judgment of hierarchy correctness/depth, precision of concept-to-category assignment against a domain taxonomy (e.g., ACM CCS), sampled manual review of extracted relationships, and a small user study comparing the CCC/knowledge-graph view against a flat acronym-list baseline.

Status: Tier 1 evaluation against the SciAI/SciAD gold benchmarks was **not carried out** in the implementation stage (Not Available — no SciAI/SciAD Precision/Recall/F1 numbers appear in any project report). Tier 2’s formal user study was also **not carried out** (Not Available). What was measured is reported in Section 17, using the metrics the implementation team actually defined (concept-extraction depth, evidence traceability, hallucination rate, and pairwise similarity statistics) — these are related to, but not the same as, the Tier 1/Tier 2 plan above. This distinction is treated as a limitation in Section 20.

10 System Architecture

The proposed ten-module pipeline (Section 7) was instantiated in practice as the **HMRP** system, whose architecture is documented in `Finalreport.md`. The implemented architecture is organized into eight functional layers:

1. INPUT & INGESTION LAYER
PDF uploads (Streamlit UI) / CLI path (`run.py --demo`) / batch dataset (HMRP_Dataset)
↓
2. EXTRACTION & PARSING LAYER
PyMuPDF (`fitz`) + `pdfplumber` fallback; SHA-256 document fingerprinting (`paper_<hash[:12]>`); section extractor (Title, Abstract, Intro, Methods, Datasets, Results, Limitations, Gap)
↓
3. PREPROCESSING & NORMALIZATION
NLTK tokenizer, lemmatizer, domain stopwords removal; noise cleaner (header/footer, URL, DOI, equation/figure artifact suppression)
↓
4. MULTI-ENGINE CONCEPT & KEYWORD EXTRACTION
TF-IDF (`scikit-learn`) · YAKE (statistical n-grams) · KeyBERT (all-MiniLM-L6-v2 + MMR diversity) · `spaCy` NER (`en_core_web_sm`) · SciBERT (`allenai/scibert_scivocab_uncased`) token & sequence classification
↓
5. HIERARCHY & TAXONOMY CONSTRUCTION LAYER
Domain scoring engine (DOMAIN_TAXONOMY keyword matching) · ConceptHierarchy tree builder (parent-child branching, pruning) · Traceability engine (sentence/page verbatim grounding)
↓
6. SIMILARITY & CLUSTERING ENGINE
TF-IDF cosine similarity · Dense semantic embedding similarity (sentence-transformers + `SciBertEmbedder`) · Hybrid mean-pooled matrix
↓
7. VALIDATION & ISOLATION ENFORCER
`PaperValidator` – 8 mandatory rules (title match, author match, concept verification, hierarchy integrity, mindmap origin, embedding validation, stale-term rejection)
↓
8. VISUALIZATION, SEARCH & EXPORT LAYER
Streamlit dashboard · Sunburst / Treemap / Icicle charts · interactive knowledge graph (`NetworkX` + `PyVis` + `Plotly`) · radial mind map · similarity heatmap · fuzzy search engine · multi-format exporter

10.1 Mapping to the Proposed Ten-Module Design

Proposed Module (Sec. 7.2)	Implemented Architectural Layer
1-3. PDF Processing, Text Extraction, Preprocessing	Layers 2-3 (extraction/pdf_extractor.py, extraction/text_processor.py)
4. Abbreviation Detection	Layer 4, SciBERT token classifier (Module 4 of acronym_model.pkl)
5. Long Form Extraction	Layer 4, extraction/concept_extractor.py (technical regex, evidence grounding)
6. Disambiguation	Layer 4, SciBERT sequence classifier (Module 6 of acronym_model.pkl) — simplified relative to design
7. Concept Extraction	Layer 4/5, keyword_extractor.py + concept_extractor.py — simplified relative to design (no embedding-anchor concept taxonomy walk)
8. CCC Mapping	Not implemented as a discrete module
9. Hierarchy Generation	Layer 5, hierarchy/concept_hierarchy.py, hierarchy/hierarchy_builder.py
10. Knowledge Graph + Visualization	Layer 8, visualization/knowledge_graph.py, hierarchy_viz.py, mindmap_viz.py, heatmap_viz.py

Layer 7 (Validation & Isolation Enforcer) has no direct counterpart in the original ten-module design; it was added during implementation specifically to address the “concept hallucination and identity drift” problem identified in the HMRP problem statement (Section 2), and is a genuine, additional engineering contribution beyond the original research design.

10.2 Project Structure (as implemented)

The codebase is organized into the following top-level packages (per Finalreport.md, Section 5): app/ (Streamlit UI and pipeline orchestrator), extraction/ (PDF, text, keyword, and concept extraction), hierarchy/ (taxonomy tree construction), similarity/ (pairwise similarity engine), visualization/ (Sunburst/Treemap/Icicle, knowledge graph, heatmap, mind map), search/ (fuzzy search engine), export/ (multi-format exporter), models/ (Paper dataclass, SciBERT embedder), utils/ (logging, helpers, PaperValidator), HMRP_Dataset/ (raw PDFs, processed splits, dataset scripts), data/ (runtime uploads/exports), and tests/ (pytest suite).

11 Dataset Description

The system is evaluated on the **HMRP Dataset**, a curated 50-paper arXiv corpus, documented in `DATASET_REPORT.md` and cross-referenced in `Finalreport.md` (Sections 6–7).

11.1 Corpus Statistics

Metric	Value
Total source PDFs scanned	50 files
Valid papers processed	50 (100.0%)
Corrupted / unreadable files	0 (0.0%)
Duplicate documents removed (via SHA-256)	0
Total corpus page count	1,123 pages
Average pages per paper	22.46 (min 9, max 50)
Average word count per paper	9,233 words (461,659 words total)
Average keywords per paper	30.72
Average research concepts extracted per paper	60.00
Primary research domains	NLP, AI, Machine Learning, Computer Vision, Bioinformatics
arXiv categories represented	cs.CL, cs.IR, cs.AI, cs.LG, cs.CV, cs.SE, cs.MA

11.2 Document-Level Leakage-Free Partitioning

Splits are strictly isolated at the paper-ID level, so that no sentence or section from a test paper is used in training or corpus-wide vocabulary construction:

Total Papers: 50

- └ Train Set (70%): 35 papers (~786 pages) → `HMRP_Dataset/processed/train.json`
- └ Validation Set (14%): 7 papers (~157 pages) → `HMRP_Dataset/processed/val.json`
- └ Test Set (16%): 8 papers (~180 pages) → `HMRP_Dataset/processed/test.json`

11.3 Processed Dataset Artifacts

- `dataset_full.json` — complete structured JSON for all 50 papers.
- `train.json` / `val.json` / `test.json` — the 35/7/8-paper splits above.
- `metadata_summary.csv` — lightweight metadata table (page counts, word counts, domain tags).
- `dataset_summary.json` — corpus-level statistical summary.

11.4 Auxiliary Data Sources (Runtime and Design-Time)

Source	Location	Format	Count	Role
Runtime uploads	data/uploads/	.pdf	Dynamic	Ad-hoc interactive analysis via the Streamlit app
Domain taxonomy seeds	app/config.py (DOMAIN_TAXONOMY)	Python dict	8 domains, 80+ seed terms	Domain classification and hierarchy construction
Fine-tuned acronym model	acronym_model.py	PyTorch state dict	—	SciBERT token/sequence classification weights (Modules 4 and 6)

Note: The 50-paper HMRP corpus is distinct from the standard SciAI/SciAD benchmark datasets used by the six core reviewed papers (Section 6.1). No evidence in the project materials indicates that HMRP was evaluated against SciAI, SciAD, or SciAD-dedupe; this is marked explicitly as a limitation in Section 20.

12 Data Preprocessing

The preprocessing pipeline, as documented in DATASET_REPORT.md and Finalreport.md, proceeds as follows:

Raw ArXiv PDFs (50 files)

- SHA-256 checksum & hash deduplication
- PyMuPDF & pdfplumber text extraction
- Metadata & section parser (Title, Abstract, Methods, Results)
- Text normalization (NLTK tokenizer, lemmatizer, stopword removal)
- Multi-engine concept extraction (spaCy + KeyBERT + YAKE + SciBERT)
- Strict document-level leakage-free splitting
- Train (70%) / Validation (14%) / Test (16%)

12.1 Key Preprocessing Steps

1. **SHA-256 content hashing:** each PDF is uniquely identified (`paper_<hash[:12]>`) for immutable tracking and duplicate detection. Zero duplicates were found in the 50-paper corpus.
2. **Text normalization:** strips headers/footers, watermark artifacts, equation-formatting noise, and invalid Unicode control characters; case is deliberately preserved (not globally lowercased) because capitalization is a primary acronym-detection signal.
3. **Structured section parsing:** automatically identifies problem statement, objective, proposed methods, datasets, limitations, and future work sections.
4. **Evidence grounding:** every extracted concept is cross-referenced with a verbatim occurrence in the source text and assigned a page number and context sentence — this is the mechanism that yields the 100% traceability figure reported in Section 17.

12.2 Zero-Leakage Assurance

Documents from the same paper/source never cross train/evaluation split boundaries, and no sentence or section from a test paper contributes to training or corpus-wide vocabulary (TF-IDF) construction. This directly targets a limitation the project’s own literature review identified in the SciAD benchmark (duplicate/conflicting labels; see Egan & Bohannon, 2021, Section 6.1, Paper 6).

13 Model / Algorithm Details

Four categories of models/algorithms are used in the implemented system.

Component	Model / Algorithm	Input	Output	Training Status
Acronym Identification (Module 4)	BertForTokenClassification - lenai/scibert_scivocab_uncased	Sentence for scientific sentences (max length 512)	BIO tags: {0: '0', 1: 'B-ABBR', 2: 'I-ABBR'}	Fine-tuned (acronym_model.pkl)
Sequence Classifier (Module 6)	BertForSequenceClassification - lenai/scibert_scivocab_uncased	Scientific token IDs	2-class probability distribution (P(Class 0), P(Class 1))	Fine-tuned (acronym_model.pkl)
Keyword Ranker	KeyBERT (all-MiniLM-L6-v2 embeddings + MMR)	Cleaned document text	Ranked n-grams with diversity-adjusted scores	Pre-trained (not further fine-tuned)
Lexical Ranker	scikit-learn TF-IDF Vectorizer	Lemmatized corpus vocabulary	TF-IDF weights	Fit dynamically on the corpus
Statistical Keyword Extractor	YAKE	Cleaned document text	Statistical n-gram keyword candidates	Unsupervised, rule/statistics-based
Named Entity Recognizer	spaCy en_core_web_sm	Text blocks ($\leq 100k$ characters)	Entity labels (ORG, PRODUCT, etc.)	Pre-trained
Semantic Embedder (similarity)	sentence-transformers/all-MiniLM-L6-v2	Document text chunks	384-dimensional dense vectors	Pre-trained
Domain/Concept Embedder	SciBERTEmbedder (allenai/scibert_scivocab_masked_mean-pooling)	Concept and domain strings	768-dimensional dense vectors	Pre-trained

13.1 Acronym Model Detail

The fine-tuned acronym model (acronym_model.pkl) bundles two SciBERT-based heads, referred to internally as **Module 4** and **Module 6**, consistent with the module numbering in the proposed ten-module design (Section 7.2):

- **Module 4 (Token Classifier):** performs BIO-tag token classification for abbreviation span detection. Per MODEL_ACCURACY_REPORT.md, this module is described

as “highly sensitive to sub-word prefixes (##), extracting domain shortforms accurately when capitalized in isolated sentences.”

- **Module 6 (Sequence Classifier):** performs binary sequence classification, reported to have a mean output probability of $P(\text{Class 1}) \approx 0.35\text{--}0.48$ across sampled test-partition sentences. This is a **simplified realization** of the “Disambiguation” step envisioned in Section 7.3: it classifies whether a sentence is scientific/acronym-context-relevant, rather than selecting among multiple explicit candidate long forms for an ambiguous acronym as the hdBERT-style design specified. No evidence in the project materials shows Module 6 performing multi-candidate argmax disambiguation.

13.2 Similarity Engine Detail

`similarity/similarity_engine.py` computes two similarity matrices and a hybrid mean-pooled combination:

1. **TF-IDF cosine similarity** — lexical overlap, from the corpus vocabulary fit in preprocessing.
2. **Dense semantic similarity** — cosine similarity of all-MiniLM-L6-v2 (384-d) embeddings, and separately of SciBertEmbedder (768-d) mean-pooled embeddings.

13.3 Validation Engine (PaperValidator)

`utils/validator.py` enforces 8 mandatory rules on every processed paper: title match, author match, concept verification, hierarchy integrity, mindmap origin check, embedding validation, and stale-term rejection (exact rule count and names per `Finalreport.md`, Sections 4 and 15). This module is the mechanism credited with the reported 0.0% hallucination rate (Section 17).

14 Proposed System / Implementation

14.1 Inference (End-to-End) Pipeline

When a user submits a new PDF via the Streamlit UI or CLI, the system executes the following nine-step pipeline (per `Finalreport.md`, Section 10):

1. **Upload** — PDF saved to `data/uploads/`.
2. **Ingestion & extraction** — SHA-256 hashing; PyMuPDF extracts text, title, authors, year, sections.
3. **Preprocessing** — noise cleaning, Unicode normalization, tokenization, lemmatization.
4. **Multi-engine keyword extraction** — TF-IDF, KeyBERT, YAKE merged into a unified, confidence-scored ranked list.
5. **Concept extraction** — spaCy NER plus technical regex; every concept linked to a verbatim sentence and page number.
6. **Hierarchy construction** — domain scoring against `DOMAIN_TAXONOMY`; `ConceptHierarchy` builds parent-child branches.
7. **Pairwise similarity** — TF-IDF and sentence-transformer cosine similarity computed against other loaded papers.
8. **Validation** — all 8 `PaperValidator` checks enforced.
9. **Visualization and rendering** — Streamlit tabs for Papers, Keywords, Hierarchy, Graph, Mindmap, Similarity Heatmap, Search, and Export.

14.2 Worked Example (from project documentation)

- **Input:** `2609.08589v1.pdf` uploaded via the Streamlit sidebar.
- **Fingerprinting:** `PDFExtractor` assigns ID `paper_7d199ff1870e`, extracts 33 pages.
- **Normalization:** `TextProcessor` strips equation artifacts, producing 9,475 cleaned tokens.
- **Concept extraction:** `KeyBERT` extracts terms such as “task progress,” “LLM agents,” “unreliable progress bar”; `spaCy` extracts entities and technical noun chunks; `ConceptExtractor` locates verbatim occurrences (e.g., Page 1, Paragraph 2).
- **Hierarchy:** `HierarchyBuilder` matches seeds for “Artificial Intelligence” and “Natural Language Processing”; `ConceptHierarchy` branches as `AI → NLP → LLM Agents → Task Progress`.
- **Validation:** all 8 `PaperValidator` checks reported as PASSED.
- **Rendering:** Sunburst chart, knowledge graph, and similarity heatmap updated.

14.3 Implemented Feature Inventory

Per `Finalreport.md`, Section 19, the following are reported as fully implemented: recursive PDF scanning with fallback extraction; SHA-256 document hashing; structured section parsing; NLTK-based text processing; hybrid keyword extraction (TF-IDF + KeyBERT + YAKE); spaCy NER and technical acronym extraction; SciBERT acronym identification (BIO tagging) and sequence classification; automated hierarchical taxonomy construction from domain seeds; verbatim evidence

grounding (page/sentence linking); dual-engine paper similarity (TF-IDF + sentence transformers); 8-rule paper validation; an 8-tab interactive Streamlit dashboard; Sunburst/Treemap/Icicle/Mindmap/Graph/Heatmap visualizers; fuzzy substring search with HTML highlighting; multi-format export (CSV, Excel, JSON, SVG, PNG, standalone HTML); and a batch dataset preprocessing/splitting pipeline.

Partially implemented: table structure parsing (text extracted; cell bounding-box geometry not exported).

Not implemented (explicitly out of the current build, per project documentation): multi-lingual parsing; distributed vector-database integration; automatic citation-network resolution; the CCC (Concept-Category-Context) chain-construction module (Module 8, Section 7.2); typed semantic-relationship extraction between concepts (Module 8b, Section 7.3); and formal multi-candidate acronym disambiguation as specified in the original design (Module 6, Section 7.3).

15 Experimental Setup

- **Corpus:** the 50-paper HMRP Dataset (Section 9), with the document-level 70/14/16 train/validation/test split.
- **Evaluation partition:** the dedicated 8-paper test split (HMRP_Dataset/processed/test.json), used for the accuracy and similarity benchmarks reported in MODEL_ACCURACY_REPORT.md.
- **Baseline comparator:** a “basic frequency/unigram” pipeline, used throughout MODEL_ACCURACY_REPORT.md as the point of comparison for the enhanced multi-engine pipeline.
- **Reproducibility artifacts:** `python3 HMRP_Dataset/process_dataset.py` (dataset construction) and `python3 HMRP_Dataset/evaluate_accuracy.py` (accuracy benchmarking), per MODEL_ACCURACY_REPORT.md, Section 7.
- **Automated test suite:** `tests/test_paper_isolation.py`, `tests/test_pipeline.py`, `tests/test_scibert.py`, run via `pytest`.
- **Hardware / compute environment:** *Not Available* — no GPU/CPU specification, batch size, or training-run configuration is reported in the project documentation for the SciBERT fine-tuning step itself (only inference-time performance figures are given; see Section 17.5).
- **Random seed:** SEED = 42, per Finalreport.md, Section 14 (reproducibility item).

16 Results & Evaluation

All figures in this section are taken directly from `MODEL_ACCURACY_REPORT.md` and `Finalreport.md`; no figures have been recalculated or estimated.

16.1 Baseline vs. Enhanced Pipeline — Accuracy and Quality Comparison

Metric / Dimension	Baseline (Unigram/Frequency)	Enhanced HMRP (Multi-Engine Hybrid)	Relative Change
Concepts extracted per paper	15.0	60.0	+300.0% depth
Multi-engine extraction	Single (word counts)	Hybrid (KeyBERT + spaCy NER + SciBERT + YAKE)	Multi-modal NLP
Domain taxonomy hierarchy	Disabled (flat list)	Automated Sunburst & Treemap	Full taxonomy mapping
Source evidence traceability	0.0% (no sentence/page link)	100.0% (page & sentence linked)	+100.0% traceability
Hallucination / stale-data rate	12.5%	0.0% (enforced by PaperValidator)	100% elimination
Semantic extraction depth	Surface unigram noise	Contextual embeddings & noun chunks	Higher semantic specificity
Hierarchy taxonomy coverage	0.0%	8 broad domains, 40+ sub-branches	New capability
Zero-leakage compliance	Not enforced	100.0% verified at paper-ID level	New capability

16.2 Acronym / Technical-Entity Benchmark (Test Partition)

Per `MODEL_ACCURACY_REPORT.md`, Section 3, evaluation of the SciBERT acronym model on unseen test-split papers reported:

- True positives (exact acronym matches): **5**
- Verified domain terms/acronyms sampled: GPT-2, UMAP, AAI, ORCID, ANR-10
- False-discovery control: stopwords suppression applied for 2-letter tokens (WE, IN, AN, FOR)

Important caveat (stated explicitly): this is a small, sampled qualitative check (5 true positives on the test partition), not a full Precision/Recall/Macro-F1 benchmark comparable to the SciAI/SciAD figures reported for Papers 1–6 (Section 6.1). No test-set-wide Precision, Recall, or F1 number for acronym identification or disambiguation is reported anywhere in the project materials — this is marked **Not Available** and discussed further in Section 18.

16.3 Semantic Similarity and Clustering Evaluation (Test Partition)

Computed with the dual engine (TF-IDF + sentence-transformers/all-MiniLM-L6-v2):

Metric	Score	Description
Mean pairwise similarity	0.2424	Distinct separation between diverse topics
Standard deviation	0.0667	Well-distributed cosine-distance space
Max pairwise similarity	0.4008	High alignment between related agent/LLM papers
Min pairwise similarity	0.1093	Strong orthogonality between distant domains

Top similar test-paper pairs:

1. 2609.08589v1 (*Progress Bars in LLM Agents*) ↔ 2609.09090v1 (*LLM Sycophancy under Pressure*) — cosine 0.4008
2. 2609.08689v1 (*Literary Periodization & LLMs*) ↔ 2609.08459v1 (*Political Authorship Stylometry*) — cosine 0.3537
3. 2609.08589v1 (*LLM Agent Progress*) ↔ 2609.08698v1 (*Record Grouping in Language Models*) — cosine 0.3535

16.4 Sample Input/Output Comparison

Test sample: arXiv:2609.08459v1 (*Detecting Authorship in Political Texts with Inductive Stylometry*)

- **Baseline extraction:** ["political", "text", "authorship", "model", "analysis"] (generic unigrams).
- **HMRP enhanced output:**
 - Proposed method: Inductive Stylometry for Speech & Text Authorship Attribution
 - Technical terms: ["UMAP", "Cluster Analysis", "Stylometric Vectors", "Cosine Distance"]
 - Identified domain: Natural Language Processing & Computational Linguistics
 - Hierarchical path: AI / NLP → Stylometry → UMAP Clustering → Authorship Verification
 - Evidence grounding: Page 2, Paragraph 3.

16.5 Automated Test Suite Results

Test Module	Result
tests/test_paper_isolation.py	7/7 PASSED
tests/test_scibert.py	1/1 PASSED

Test Module	Result
tests/test_pipeline.py	14/16 PASSED
Total	22/24 PASSED

Two test cases in test_pipeline.py are reported as failing; no root-cause detail for the two failures is given in the project documentation (marked **Not Available**).

16.6 Performance (Inference-Time) Measurements

Operation	Measurement
PDF ingestion & extraction	~0.8-1.5 s per 20-page PDF
KeyBERT + YAKE extraction	~1.2 s per paper
SciBERT acronym inference	~0.05 s per sentence (CPU/MPS)
Pairwise similarity matrix (50×50 papers)	< 0.2 s
Peak memory footprint	~1.2 GB RAM

17 Comparative Analysis

17.1 Stage-Coverage Comparison Against the Reviewed Literature

Extending the stage-mapping table from the *Research Blueprint* (Section 3, Step 3.8 of the design phase) with a column for what was actually built:

Pipeline Stage	Covered by Reviewed Literature?	Covered by Proposed Design (Sec. 7)?	Implemented in HMRP?
PDF Processing / Text Extraction	Yes (implicit, Papers 1-6)	Yes	Yes
Abbreviation Identification	Yes (Papers 1, 2, 6)	Yes	Yes (SciBERT BIO tagging)
Long Form Extraction	Yes (Papers 1, 3)	Yes	Yes (rule-based/regex)
Disambiguation (multi-candidate)	Yes (Papers 1, 3, 4, 5, 6)	Yes (hdBERT-style)	Partial (binary sequence classifier only)
Concept Identification	Not covered	Yes (novel)	Partial (domain-keyword matching, not embedding-anchor taxonomy walk)
CCC Mapping	Not covered	Yes (core novel contribution)	No
Hierarchy Generation	Not covered	Yes	Yes (domain-taxonomy tree, not CCC-chain-derived)
Knowledge Graph	Not covered	Yes	Yes (NetworkX-based, entities/domains as nodes)
Typed Semantic Relationships	Not covered	Yes (pattern + fallback classifier)	No
Interactive Visualization	Not covered (MadDog: lookup only)	Yes	Yes (Sunburst, Treemap, Icicle, Mindmap, Graph, Heatmap)

This table shows that the implementation closes the *interactive-visualization* and *hierarchy* portions of the identified research gap (Section 5) in a generalized, domain-taxonomy-driven form, but does **not** close the specific CCC-mapping / typed-relationship-extraction portion of the gap as originally scoped in Modules 8 and 8b.

17.2 Quantitative Comparability to Papers 1-6

A direct numerical comparison between HMRP’s acronym-related components and the Macro-F1 scores reported by Papers 1-6 (Section 6.1, range 81.90-94.12 F1 on SciAI/SciAD) is **not possible with the evidence available**: HMRP’s acronym model was evaluated on a small, non-standard sample (5 true positives on an 8-paper test split; Section 14.2), not on SciAI, SciAD, or SciAD-dedupe. Any numerical comparison would therefore be invalid and is deliberately omitted. What *can* be compared, and is compared in Section 14.1, is HMRP’s own before/after pipeline comparison (unigram baseline vs. multi-engine hybrid), which is an internally consistent, project-defined benchmark rather than a benchmark shared with the reviewed literature.

17.3 Comparison Against Clinical/Biomedical and Knowledge-Graph Themes (B, C, D)

- Compared with Theme B systems (e.g., BlueBERT at 95-98% accuracy on MSH WSD/UMN), HMRP’s SciBERT modules operate on a different domain (general AI/CS arXiv papers vs. clinical notes) and were not evaluated on a comparable disambiguation benchmark — no direct comparison is possible.
- Compared with Theme C systems (e.g., Graphusion, NLP-AKG), HMRP’s knowledge graph is smaller in scope (per-corpus, 50 papers, domain-taxonomy-seeded) rather than a large-scale, LLM-extracted academic knowledge graph (NLP-AKG: 620,353 entities). HMRP’s design choice — deterministic taxonomy/dictionary lookup rather than LLM-based triplet extraction — was explicitly motivated in the *Research Blueprint* (Section 4.2) by Graphusion’s documented hallucination risk, and is corroborated by HMRP’s own 0.0% measured hallucination rate (Section 14.1), though the two hallucination-rate figures are not measuring identical things (Graphusion’s is about triplet correctness; HMRP’s is about PaperValidator-caught cross-document contamination).

18 Discussion

The project’s two stages tell a coherent but only partially convergent story. The research-design stage correctly identifies, through a systematic review of 6 and then 22 papers, that the field’s unmet need is not further incremental gains in Acronym Identification/Disambiguation accuracy (which had already reached 91-94% F1 by 2021) but a structured, evidence-grounded, and visually explorable layer on top of resolved acronyms. The implementation stage responds to this need, but does so by generalizing the target: rather than building a strict, per-acronym CCC (Concept-Category-Context) chain with typed semantic edges, HMRP builds a **domain-level** hierarchy and knowledge graph driven by a fixed taxonomy (DOMAIN_TAXONOMY) and a multi-engine keyword/concept extractor, with every node traceable to a verbatim sentence and page (satisfying the “Context” component of CCC even though the “Category” component is domain-level rather than acronym-specific).

This generalization is a reasonable engineering trade-off: it sidesteps the harder, still partially unsolved sub-problem of multi-candidate acronym disambiguation (Papers 4 and 5 in Section 6.1 report this as their hardest residual error class — near-synonymous candidates and context-insufficient sentences) and instead delivers a system that is measurably better than a naive baseline (300% more concepts per paper, 100% traceability, 0% hallucination) on the metrics the project team chose to track. The trade-off’s cost is that the implemented system’s contribution is best framed as “multi-engine, evidence-grounded literature-mapping for a paper corpus” rather than “acronym-to-CCC-hierarchy resolution” as originally scoped — a narrower and, evidentially, better-supported claim.

The most significant unresolved item is disambiguation: the design phase’s Module 6 specifies choosing among multiple *named* candidate long forms for an ambiguous acronym (e.g., “AD” → *Acronym Disambiguation* vs. *Alzheimer’s Disease* vs. *Anno Domini*); the implemented Module 6 is a binary sequence classifier whose output ($P(\text{Class } 1) \approx 0.35\text{-}0.48$) is not documented as being wired to a candidate-selection step. This is consistent with the project’s own MODEL_ACCURACY_REPORT.md, which never states an AD accuracy figure, in contrast to its explicit AI-related figures.

19 Limitations

The following limitations are drawn directly from the project’s own documentation (`Finalreport.md`, Section 23) and supplemented with limitations that follow from the design-vs-implementation gap established in Sections 7–15 of this paper.

Documented in the project’s own materials:

1. **Scanned/image PDFs:** PDFs without an embedded text layer require an OCR preprocessing step (Tesseract/pdf2image) not currently integrated into the main pipeline.
2. **Single-node, in-memory design:** pairwise similarity is computed in-memory with NumPy; corpora beyond roughly 5,000 papers would require an external approximate-nearest-neighbor vector index.
3. **English-only:** stopwords lists and domain taxonomies are tailored to English academic literature.

Identified through this paper’s cross-check of design vs. implementation:

4. **No standard-benchmark evaluation of acronym identification/disambiguation:** the project reports no Precision/Recall/Macro-F1 figures on SciAI, SciAD, or SciAD-dedupe, so the acronym-related components cannot be positioned on the same scale as Papers 1–6 (Section 15.2).
5. **CCC mapping and typed relationship extraction are unimplemented:** the “core proposed contribution” of the design phase (Module 8, Section 7.3) has no corresponding code module in the documented project structure.
6. **Multi-candidate disambiguation is unimplemented:** Module 6, as built, is a binary relevance/sequence classifier, not a candidate-selection disambiguator (Section 11.1).
7. **Small, non-standard acronym-evaluation sample:** the only acronym-specific accuracy evidence is a 5-true-positive qualitative check on an 8-paper test split (Section 14.2), which is not statistically representative.
8. **Two of 24 automated tests fail,** with no documented root cause (Section 14.5).
9. **No hardware/training configuration is documented** for the SciBERT fine-tuning step that produced `acronym_model.pkl` (Section 13), limiting reproducibility of that specific artifact.
10. **The Tier-2 evaluation plan (expert/human judgment of hierarchy quality, CCC-precision spot-checks, relationship-correctness review, and a user study; Section 7.5) was never executed,** so the *usability* benefit of the implemented hierarchy/graph/visualization layer over a flat list — the central motivating claim of the whole project — is asserted in the design documents but not empirically demonstrated in the implementation documents.

20 Future Work

Future work falls into two categories: work already identified by the project team, and work implied by the gaps documented in this paper.

20.1 Already Identified by the Project Team

- **OCR ingestion pipeline** for scanned/historical paper archives.
- **Dynamic knowledge-graph merging** across papers via external citation APIs (e.g., OpenAlex, Semantic Scholar).
- **Cross-lingual embedding models** (e.g., sentence-transformers/paraphrase-multilingual-mpnet-base-v2) for multilingual scientific text.
- **Distributed vector-database integration** (e.g., Milvus/Qdrant) for corpora beyond ~100,000 papers.
- **Automatic citation-network resolution** via the CrossRef API.

20.2 Implied by the Design-vs-Implementation Gap (This Paper’s Contribution)

- **Implement the CCC Mapping module (Module 8)** as originally specified: a recursive, per-concept chain from resolved acronym $\rightarrow$ concept $\rightarrow$ category $\rightarrow$ root domain, with sentence/section provenance retained at each link, rather than the current domain-level hierarchy alone.
- **Implement multi-candidate acronym disambiguation** (Module 6 as designed): either an hdBERT-style dual-path fusion model or a retrieval-based Siamese approach, evaluated against explicit candidate long-form sets.
- **Implement typed relationship extraction (Module 8b)** using the pattern-matching-plus-fallback-classifier design already specified (`used_for`, `belongs_to`, `part_of`, `extends`, `improves`, `depends_on`, `similar_to`), to move the knowledge graph from an entity/domain graph to a semantically typed graph.
- **Run the Tier-1 evaluation** (Precision/Recall/Macro-F1 on SciAI, SciAD, and SciAD-dedupe) to make the acronym-related components directly comparable to Papers 1–6.
- **Run the Tier-2 evaluation** (expert hierarchy-quality review, CCC-precision spot-check against ACM CCS, relationship-correctness review, and the planned user study) to empirically substantiate the usability claims currently only asserted in the design documents.
- **Diagnose and fix the two failing test_pipeline.py cases**, and document the training configuration used to produce `acronym_model.pkl`, to close the reproducibility gaps identified in Section 17.
- **Extend beyond the AI/CS domain**, following the multi-domain feasibility already demonstrated by MadDog (Paper 3, Section 6.1) and the clinical-domain literature reviewed in Theme B (Section 6.2), once the AI/CS-domain pipeline is fully validated against Tier 1 and Tier 2.

21 Conclusion

This paper has presented a single, evidence-based account of a two-stage research project: a systematic literature review and gap analysis (6 papers, extended to 22 across four themes) that motivates a ten-module CCC-mapping pipeline for scientific acronym understanding, and a working prototype (HMRP) that implements a substantial, but not complete, subset of that pipeline. The prototype demonstrably improves concept-extraction depth (+300%), evidence traceability (0% $\rightarrow$ 100%), and hallucination avoidance (12.5% $\rightarrow$ 0%) over a naive unigram baseline on a 50-paper, leakage-free arXiv corpus, and provides an interactive, multi-format visualization and export layer that directly addresses the “no interactive visualization” gap identified in the literature review. At the same time, the prototype does not implement the specific CCC chain-construction algorithm, typed semantic-relationship extraction, or multi-candidate acronym disambiguation that the design phase identified as its core novel contribution, and it has not been benchmarked against the SciAI/SciAD gold-standard datasets used throughout the reviewed literature. The research gap identified in Phases 1–3 therefore remains only partially closed: the *hierarchy and visualization* half of the gap has working, measured evidence behind it; the *CCC mapping, disambiguation, and typed-relationship* half remains a well-specified but unimplemented direction for future work. Reporting both facts together, rather than either alone, is the central contribution of this paper.

22 References

References are listed exactly as documented in the project’s literature-review materials (Phase1_Problem_Understanding.md, Phase2_Literature_Review.md, Paper_summary.pdf, and Consolidated_Paper_Comparison.xlsx). No reference has been added, removed, or altered beyond formatting.

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